Linear Algebra – MAT 2610

Section 2.9 (Dimension and Rank)

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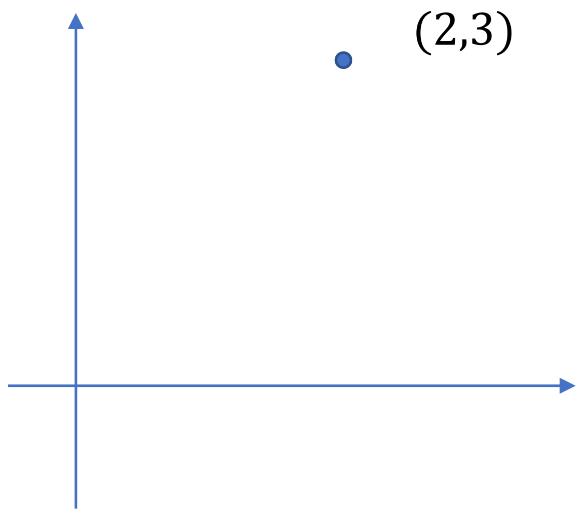
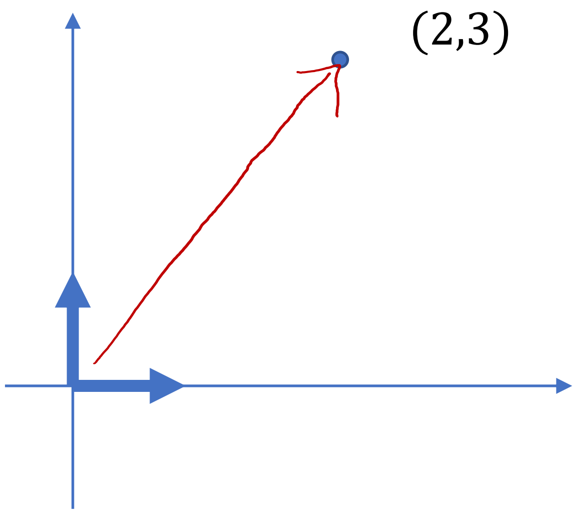
Consider a point on the plane (a, b). We can plot this point, such as $(2,3)$



What does $(2,3)$ really mean? It means “go two units in the x-direction, and 3 units in the y-direction”.

Think about this in terms of a basis for R^2

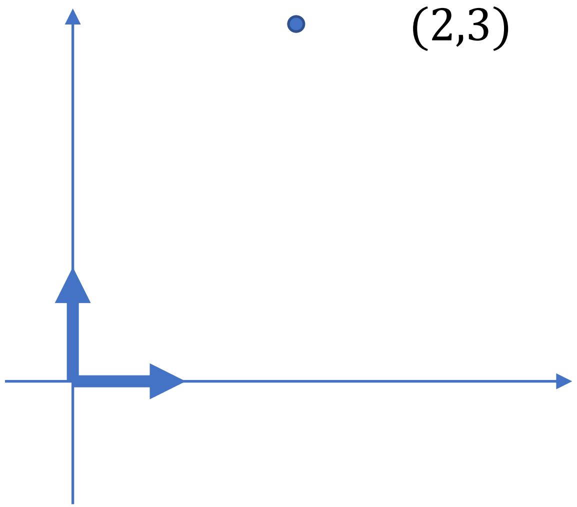
Our standard basis for R^2 is $\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. These vectors can be interpreted as a vector of length 1 in the x-direction and a vector of length 1 in the y-direction.



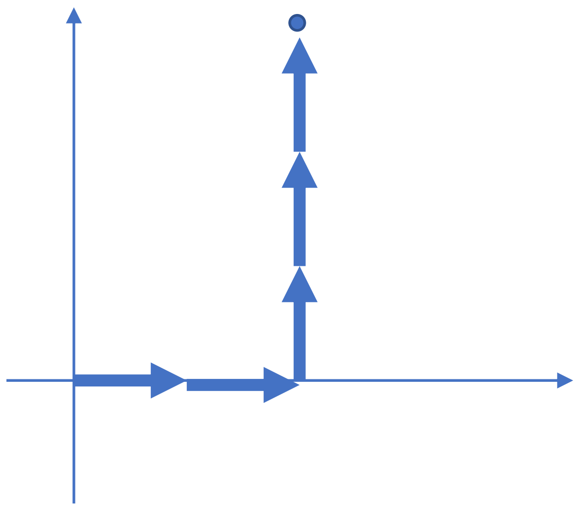
When we introduct this concept, then $(2,3)$ means "two units in the direction of the first vector, then 3 units in the direction of the second vector". We will often describe something like $(2,3)$ as the coordinates of a point, but in reality we should think of this as the **coordinates in relation to the basis** $\boldsymbol{\mathcal{S}} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$.

But what is special about $\mathcal{S} = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$? It is convenient, easy to picture, and we are used to it. But mathematically, there is nothing special about it. It is just one of infinitely many possible bases.

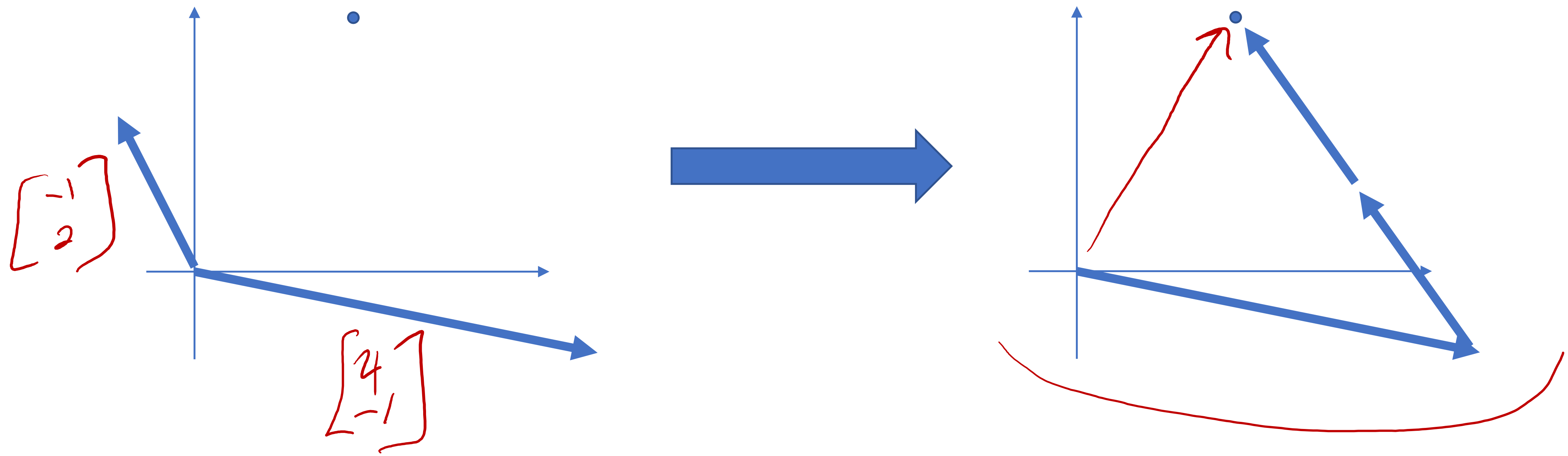
Any point on R^2 can be described in terms of any basis of R^2 .



For example, with the basis $\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ we can describe our point as a vector $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$.
This can be thought of as a description of how to reach the point we are interested in: 2 times the first vector, then three times the second vector gets us to the point on the plane in which we are interested.



But what if we take a different basis, such as, $\mathcal{B} = \left\{ \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$? We can reach the same point using this basis: One times the first vector, followed by two times the second vector brings me to the same point.



Now I need to describe this a bit differently for our basis $\mathcal{B} = \left\{ \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$, and this is where we use the **coordinates in relation to a basis** are used. For this example, we say that this point is at the coordinates $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in relation to the basis $\mathcal{B} = \left\{ \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$.

$v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

We write this as $[v]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

This means our point is $(1)\begin{bmatrix} 4 \\ -1 \end{bmatrix} + (2)\begin{bmatrix} -1 \\ 2 \end{bmatrix}$, or $\begin{bmatrix} 4 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

We have seen that $\begin{bmatrix}2\\3\end{bmatrix}$ is really just the coordinates of the same point in relation to the standard basis $\mathcal{S} = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$. So this can be written as $[v]_{\mathcal{S}} = \begin{bmatrix}2\\3\end{bmatrix}$.

$$\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$$

We are accustomed to thinking of coordinates in relation to the x-axis and y-axis, which are the standard vectors. In reality, coordinates change when you change your basis.

When we write $\begin{bmatrix}4 & -1\\-1 & 2\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$, we are taking the coordinates using one basis $[v]_{\mathcal{B}}$ and changing them into the coordinates using a different basis $[v]_{\mathcal{S}}$.

Note: Let $\mathcal{B} = \{b_1, b_2, \dots, b_k\}$ be a basis for a subspace V of R^n. Any vector v in V can be uniquely represented as a linear combination of the vectors in $\mathcal{B}$; i.e.: there are unique scalars $\{a_1, a_2, \dots, a_k\}$ such that $v = a_1 b_1 + \dots + a_k b_k$. This is only possible if the vectors in the basis are linearly independent (which they are by definition).

Example: Let V be a subspace with basis $\mathcal{B} = \left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \right\}$. Write $v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ as a linear combination of the vectors in $\mathcal{B}$

ie: what are coords of v in $\mathcal{B}$, or $[v]_{\mathcal{B}}$

$$\begin{bmatrix} 0 & 0 & -1 & 1 \\ 0 & 4 & 2 & 2 \\ -1 & 4 & 2 & 3 \end{bmatrix} \xrightarrow{r_3 \leftrightarrow r_1} \begin{bmatrix} -1 & 4 & 2 & 3 \\ 0 & 4 & 2 & 2 \\ 0 & 0 & -1 & 1 \end{bmatrix} \xrightarrow{r_1 - r_2 \to r_1} \begin{bmatrix} -1 & 0 & 0 & 1 \\ 0 & 4 & 2 & 2 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow{r_2 + 2r_3 \to r_2} \begin{bmatrix} -1 & 0 & 0 & 1 \\ 0 & 4 & 0 & 4 \\ 0 & 0 & -1 & 1 \end{bmatrix} \xrightarrow[\substack{\frac{1}{4} r_2 \to r_2 \\ -r_3 \to r_3}]{-r_1 \to r_1} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \qquad [v]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

Definition: Let $\mathcal{B} = \{b_1, b_2, \dots, b_k\}$ be a basis for R^n. For any vector v in R^n there are unique scalars $\{c_1, c_2, \dots, c_k\}$ such that $v = c_1 b_1 + \cdots + c_k b_k$. The vector

$$\begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_k \end{bmatrix}$$

is called the **coordinate vector** of v relative to B, or the $\mathcal{B}$**-coordinate vector**. We denote the $\mathcal{B}$-coordinate vector of v as $[v]_B$

Example: Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix} \right\}$. Find v if $[v]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 & 3 \\ 1 & 3 & 5 \end{bmatrix} \xrightarrow{r_1 - r_2 \to r_2} \begin{bmatrix} 1 & -1 & 3 \\ 0 & -4 & -2 \end{bmatrix} \xrightarrow{-\frac{1}{4} r_2 \to r_2} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$$

$$\xrightarrow{r_1 + r_2 \to r_1} \begin{bmatrix} 1 & 0 & 7/2 \\ 0 & 1 & 1/2 \end{bmatrix}$$

$$[v]_{\mathcal{B}} = \begin{bmatrix} 7/2 \\ 1/2 \end{bmatrix}$$

Example: Let $\mathcal{B} = \left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\2\end{bmatrix}\right\}$ a basis of R^3. Find $[v]_\mathcal{B}$ if $v = \begin{bmatrix}0\\0\\0\end{bmatrix}$

$$[v]_{\mathcal{B}} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

Dimension of a Subspace

Fact: Let V be a non-zero subspace of R^n. Then any two bases for V contain the same number of vectors

A basis for a subspace is the smallest possible generating set

The size of any basis for V is the **dimension** of V, denoted $\dim(V)$

The **rank** of a matrix A, denoted by $rank(A)$, is the dimension of the column space of A.

Example: Find a basis for the subspace V $= \left\{ \begin{bmatrix} 6r+t \\ r-s+4t \\ 5s+9t \\ 4r-5s+t \end{bmatrix} \in R^4 : r, s, t \text{ scalars} \right\}$.

Note that this is equal to

$$r\underline{\begin{bmatrix} 6 \\ 1 \\ 0 \\ 4 \end{bmatrix}} + s\underline{\begin{bmatrix} 0 \\ -1 \\ 5 \\ -5 \end{bmatrix}} + t\underline{\begin{bmatrix} 1 \\ 4 \\ 9 \\ 1 \end{bmatrix}}$$

$$V = \text{Span}\left\{ \begin{bmatrix} 6 \\ 1 \\ 0 \\ 4 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 5 \\ -5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 9 \\ 1 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 6 & 0 & 1 \\ 1 & -1 & 4 \\ 0 & 5 & 9 \\ 4 & -5 & 1 \end{bmatrix} \longrightarrow \longrightarrow \longrightarrow \begin{bmatrix} \boxed{6} & 0 & 1 \\ 0 & \boxed{5} & 9 \\ 0 & 0 & \boxed{1} \\ 0 & 0 & 0 \end{bmatrix}$$

Three pivot cols $\longrightarrow$ lin ind set

<u>or</u> col space basis is all 3 columns

$$\Rightarrow \text{The basis is } \left\{ \begin{bmatrix} 6 \\ 1 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 5 \\ -5 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 9 \\ 1 \end{bmatrix} \right\}$$

Definition: The number of vectors in a basis for a nonzero subspace V of R^n is called the **dimension** of V. We say the dimension of the zero subspace is 0.

Example: R^2 has dimension 2.

Example: R^n has dimension n.

Example: We saw that $V = \left\{ \begin{bmatrix} 6r+t \\ r-s+4t \\ 5s+9 \\ 4r-5s+t \end{bmatrix} \in R^4 : r, s, t \ scalars \right\}$ has dimension 3

Example: What is $\dim(V) = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} \in R^5 : v_1 = v_3 = 0 \right\}$ $= \left\{ \begin{bmatrix} 0 \\ v_2 \\ 0 \\ v_4 \\ v_5 \end{bmatrix} \right\}$

$$= v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + v_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + v_5 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

So $\{e_2, e_4, e_5\}$ is a gen set for V. We can see $\{e_2, e_4, e_5\}$ are lin ind, so $\{e_2\ e_4\ e_5\}$ is a basis for V, so $\dim(V) = 3$

Theorem 14: If a matrix A has n columns, then $rank(A) + \dim(Null(A)) = n$.

Theorem 15 (The Basis Theorem): Let H be a p −dimensional subspace of R^n. Any linearly independent set of exactly p elements in H is automatically a basis for H. Also, any set of p elements of H that spans H is automatically a basis for H.

Ex: $\mathbb{R}^5$

any 5 vectors from $\mathbb{R}^5$ that are lin ind are automatically a basis

Example: Let $A = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \end{bmatrix}$. Find a basis for the column space and null space.

What are their dimensions?

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \end{bmatrix} \xrightarrow{r_2 - r_1 \to r_2} \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 2 & 2 & 2 & 4 \end{bmatrix} \xrightarrow{r_3 - 2r_1 \to r_3} \begin{bmatrix} \boxed{1} & 1 & 1 & 2 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So $\dim(\mathrm{col}(A)) = \mathrm{rank}(A) = 2$

$$\text{Basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 1 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 2 & 0 \\ 2 & 2 & 2 & 4 & 0 \end{bmatrix} \xrightarrow{r_2 - r_1 \to r_2} \begin{bmatrix} 1 & 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & 4 & 0 \end{bmatrix} \xrightarrow{r_3 - 2r_2 \to r_3} \begin{bmatrix} 1 & 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{r_1 - r_2 \to r_1} \begin{bmatrix} \boxed{1} & 0 & 1 & 1 & 0 \\ 0 & \boxed{1} & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 + x_3 + x_4 = 0$

$x_2 + x_4 = 0$

$\left.\begin{matrix} x_3 \\ x_4 \end{matrix}\right\}$ free

$x_1 = -x_3 - x_4$

$x_2 = -x_4$

$x_3 = x_3$

$x_4 = x_4$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\} \qquad \dim(\mathrm{Nul}(A)) = 2$$

Theorem 8 (Section 2.3)

Invertible Matrix Theorem: If A is a square $n \times n$ matrix, then the following are equivalent:

a. The matrix A is invertible
b. The matrix A is row equivalent to I_n
c. There are n pivot positions in A
d. The equation $Ax = 0$ has only the trivial solution
e. The columns of A are linearly independent
f. The linear transformation $x \longmapsto Ax$ is one-to-one
g. The equation $Ax = b$ has at least one solution for every b in R^n
h. The columns of A span R^n
i. The linear transformation $x \longmapsto Ax$ maps R^n onto R^n
j. There is an $n \times n$ matrix C such that $AC = CA = I_n$
k. The matrix A^T is invertible

Invertible Matrix Theorem (continued)

If A is a square $n \times n$ matrix, then the following are equivalent to the statement that A is an invertible matrix:

l. The columns of A form a basis of R^n

$m.$ $Col(A) = R^n$

$n.$ $\dim(Col(A)) = n$

$o.$ $rank(A) = n$

$p.$ $Nul(A) = \{0\}$

$q.$ $\dim(Nul(A)) = 0$

Steps to show that a set B is a basis for a subspace V:

1. Show that every vector in B is also in V
2. Show that B is either linearly independent, or is a generating set of V
3. Compute the dimension of V and verify that the number of vectors in B is equal to the dimension of V

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